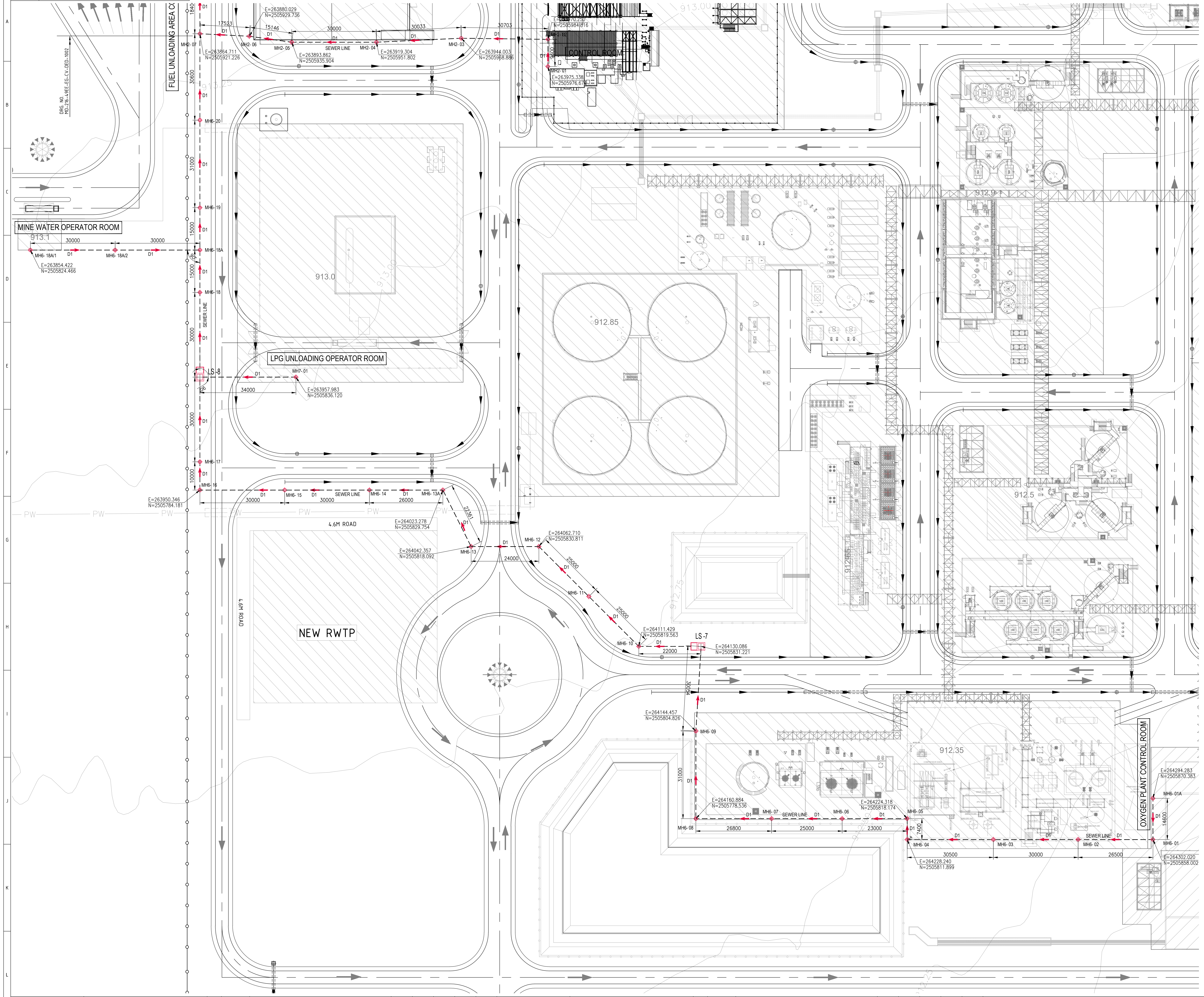
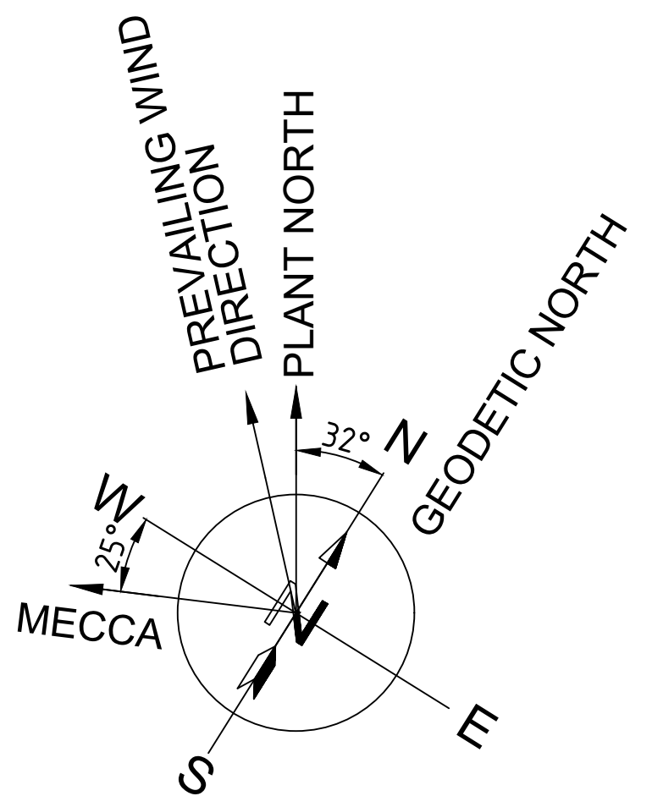
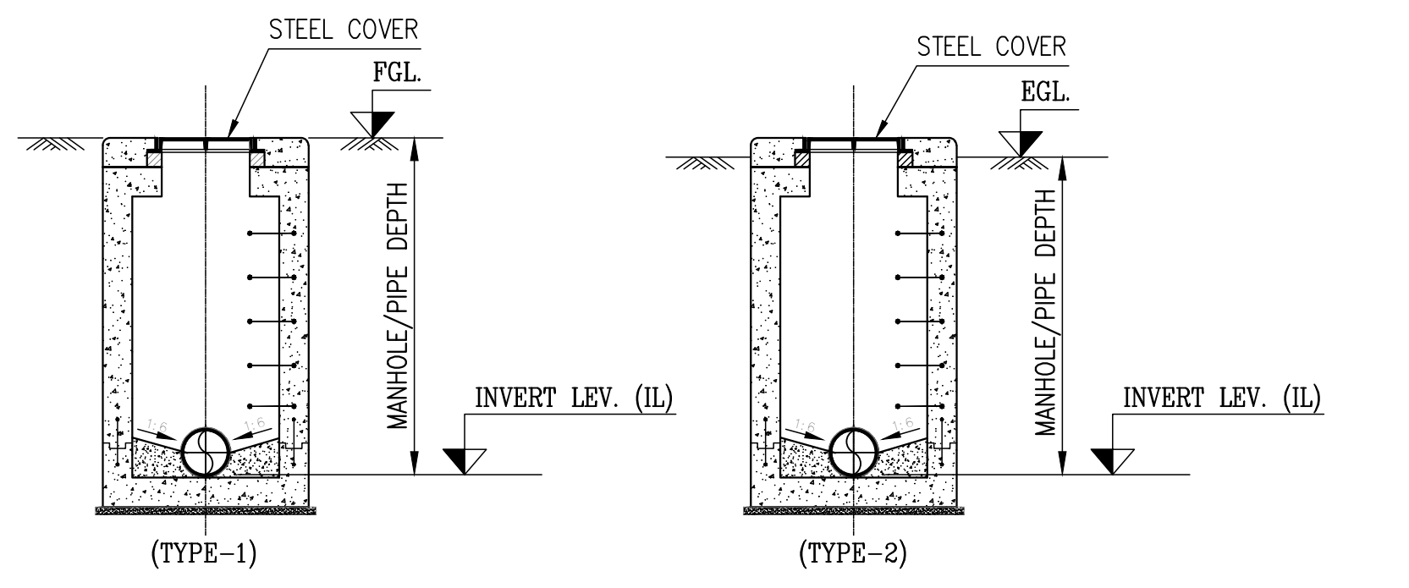




## SCHEDULE OF SEWERAGE NETWORK MANHOLES :-

| SL. NO. | SEWERAGE MANHOLE | TOP OF MANHOLE COVER LEVEL (M) | ENTRY INVERT LEVEL (M) | EXIT INVERT LEVEL (M) | CONSIDERING EGL. OR FGL. (M) | PIPE (EXIT) DEPTH FROM EGL. OR FGL. (M) | MANHOLE INTERNAL PLAN DIMENSIONS (MM) | CLEAR COVER OPENING DIMENSIONS (MM) | TYPE OF SOLID COVER | CO-ORDINATES AT CENTER OF MANHOLE | | TYPE OF MANHOLE |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | | E (M) | N (M) | |
| 1 | MH6- 01A | 912.35 | - | 911.35 | 912.35 | 1 | Ø900 | Ø750 | Group-4 | E=264294.283 | N=2505870.383 | TYPE-1 |
| 2 | MH6- 01 | 912.35 | 911.205 | 911.18 | 912.35 | 1.17 | Ø900 | Ø750 | Group-4 | E=264302.020 | N=2505858.002 | TYPE-1 |
| 3 | MH6- 02 | 912.35 | 910.915 | 910.89 | 912.35 | 1.46 | Ø900 | Ø750 | Group-4 | E=264279.547 | N=2505843.959 | TYPE-1 |
| 4 | MH6- 03 | 912.35 | 910.59 | 910.565 | 912.35 | 1.785 | Ø900 | Ø750 | Group-4 | E=264254.105 | N=2505828.061 | TYPE-1 |
| 5 | MH6- 04 | 912.35 | 910.26 | 910.235 | 912.35 | 2.115 | Ø1200 | Ø750 | Group-4 | E=264228.240 | N=2505811.899 | TYPE-1 |
| 6 | MH6- 05 | 912.35 | 910.16 | 910.135 | 912.35 | 2.215 | Ø1200 | Ø750 | Group-4 | E=264224.318 | N=2505818.174 | TYPE-1 |
| 7 | MH6- 06 | 912.35 | 909.905 | 909.88 | 912.35 | 2.47 | Ø1500 | Ø750 | Group-4 | E=264204.813 | N=2505805.986 | TYPE-1 |
| 8 | MH6- 07 | 912.35 | 909.63 | 909.605 | 912.35 | 2.745 | Ø1500 | Ø750 | Group-4 | E=264183.612 | N=2505792.738 | TYPE-1 |
| 9 | MH6- 08 | 912.35 | 909.335 | 909.31 | 912.35 | 3.04 | Ø1500 | Ø750 | Group-4 | E=264160.884 | N=2505778.536 | TYPE-1 |
| 10 | MH6- 09 | 912.35 | 909 | 908.975 | 912.35 | 3.375 | Ø1500 | Ø750 | Group-4 | E=264144.457 | N=2505804.826 | TYPE-1 |
| 11 | LS- 7 | 912.85 | 908.615 | 911.7 | 912.7 | - | - | - | - | - | - | - |
| 12 | MH6- 10 | 912.85 | 911.515 | 911.49 | 912.75 | 1.26 | Ø900 | Ø750 | Group-4 | E=264111.429 | N=2505819.563 | TYPE-2 |
| 13 | MH6- 11 | 912.9 | 911.24 | 911.215 | 912.8 | 1.585 | Ø900 | Ø750 | Group-4 | E=264087.069 | N=250582.517 | TYPE-2 |
| 14 | MH6- 12 | 912.9 | 910.965 | 910.94 | 912.8 | 1.86 | Ø900 | Ø750 | Group-4 | E=264062.710 | N=2505830.811 | TYPE-2 |
| 15 | MH6- 13 | 912.95 | 910.692 | 910.667 | 912.85 | 2.183 | Ø1200 | Ø750 | Group-4 | E=264042.357 | N=2505818.092 | TYPE-2 |
| 16 | MH6- 13A | 913 | 910.435 | 910.41 | 912.9 | 2.49 | Ø1200 | Ø750 | Group-4 | E=264023.278 | N=2505829.754 | TYPE-2 |
| 17 | MH6- 14 | 913.025 | 910.14 | 910.115 | 912.925 | 2.81 | Ø1500 | Ø750 | Group-4 | E=264001.228 | N=2505815.976 | TYPE-2 |
| 18 | MH6- 15 | 913.05 | 909.805 | 909.78 | 912.95 | 3.17 | Ø1500 | Ø750 | Group-4 | E=263975.787 | N=2505800.078 | TYPE-2 |
| 19 | MH6- 16 | 913.075 | 909.47 | 909.445 | 912.975 | 3.53 | Ø1500 | Ø750 | Group-4 | E=263950.346 | N=2505784.181 | TYPE-2 |
| 20 | MH6- 17 | 913.1 | 909.34 | 909.315 | 913 | 3.685 | Ø1500 | Ø750 | Group-4 | E=263945.047 | N=2505792.662 | TYPE-2 |
| 21 | MH7- 01 | 913 | - | 912 | 913 | 1 | Ø900 | Ø750 | Group-4 | E=263957.983 | N=2505836.120 | TYPE-1 |
| 22 | LS- 8 | 913.2 | 909.015 (From MH6- 17)<br>911.660 (From MH7- 01) | 912.05 | 913.05 | - | - | - | - | - | - | - |
| 23 | MH6- 18 | 913.2 | 911.785 | 911.76 | 913.1 | 1.34 | Ø900 | Ø750 | Group-4 | E=263913.252 | N=2505843.545 | TYPE-2 |
| 24 | MH6- 18A/1 | 913.35 | - | 912.25 | 913.25 | 1 | Ø900 | Ø750 | Group-4 | E=263854.422 | N=2505824.466 | TYPE-2 |
| 25 | MH6- 18A/2 | 913.25 | 911.95 | 911.925 | 913.15 | 1.225 | Ø900 | Ø750 | Group-4 | E=263879.861 | N=2505840.368 | TYPE-2 |
| 26 | MH6- 18A | 913.25 | 911.625 (From MH6-18A/2)<br>911.610 (From MH6- 18) | 911.585 | 913.15 | 1.565 | Ø900 | Ø750 | Group-4 | E=263905.303 | N=2505856.265 | TYPE-2 |
| 27 | MH6- 19 | 913.25 | 911.435 | 911.41 | 913.15 | 1.74 | Ø900 | Ø750 | Group-4 | E=263897.354 | N=2505868.986 | TYPE-2 |
| 28 | MH6- 20 | 913.3 | 911.1 | 911.075 | 913.2 | 2.125 | Ø1200 | Ø750 | Group-4 | E=263880.927 | N=2505895.276 | TYPE-2 |



TYPICAL CROSS SECTION OF MANHOLE

## NOTES:~

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND INVERT LEVEL & CO-ORDINATES ARE IN METRES UNLESS NOTED OTHERWISE.

2. INVERT LEVEL (IL) HAVE BEEN INDICATED FOR POINT OF SEWERAGE DRAIN AND UNIFORM SLOPE SHOULD BE PROVIDED BETWEEN THE MANHOLE.

3. MANHOLE (MH) LOCATIONS NEAR TOILET ARE TENTATIVE.

4. LENGTH OF SEWER WATER PIPE BETWEEN MANHOLES ARE INDICATED IN THIS DRAWING. THESE LENGTHS ARE APPROXIMATE (HORIZONTAL) AND TO BE ADJUSTED DURING CONSTRUCTION FOR PROPER FIXING.

5. SEWERAGE COLLECTION PIPE/DRAIN SHALL BE PROVIDED FROM TOILET/OTHER FACILITIES LOCATIONS UP TO MANHOLES WHEREVER REQUIRED. OTHERWISE CONNECTION PIPE/DRAIN SHALL BE PROVIDED BETWEEN S.P/W.P. AND NEAREST SEWERAGE MANHOLE.

6. THIS DRAWING IS APPLICABLE ONLY FOR SEWERAGE DRAIN NETWORK AND MANHOLE LOCATIONS. ALL OTHER INFORMATION, eg. LEVEL(EC.),FACILITY OUTLINE,GRADING ETC. ARE SHOWN IN THIS DRAWING FOR INFORMATION ONLY.

7. INVERT LEVELS (IL) HAVE BEEN INDICATED FOR THE MANHOLES AND UNIFORM SLOPE SHOULD BE PROVIDED IN THE SANITARY SEWER LINES BETWEEN THE MANHOLES.

8. PIPE BETWEEN LIFTING STATION (LS) AND NEXT MANHOLE SHALL BE PRESSURE PIPE, FOR LAYOUT AND DETAILS REFER UTILITY DRAWINGS.

9. FOR S.P./W.P. LOCATION AND DETAILS REFER ARCHITECTURAL DRAWINGS.

10. THIS DRAWING SHOULD BE READ IN CONJUNCTION WITH DRG. NO.
MD-216-49EE-EG-CV-DED-1001
MD-216-49EE-EG-CV-DED-1002
MD-216-49EE-EG-CV-DED-1003 &
MD-216-49EE-EG-CV-DED-1004.

## LEGENDS: –

LS = LIFTING STATION
EGL = EXISTING GROUND LEVEL
FGL = FINISHED GROUND LEVEL
DRG. = DRAWING
S.P. = SOIL PIT
W.P. = WASTE PIT

LIST OF DRAWING NO. OF LIFTING STATION :-

| LIFTING STATION | DRAWING NO. |
|---|---|
| LS-7 | MD-216-49EE-EG-CV-GAD-1595 |
| LS-8 | MD-216-49EE-EG-CV-GAD-1596 |

| Schedule of Sewer Water PVC Pipes / Encased PVC Pipes | | |
|---|---|---|
| Pipes | Diameter | Slope |
| D1 | 150 | 1/100 |
| D2 | 200 | 1/150 |
| D3 | 250 | 1/150 |

| DOCUMENT NUMBER | REFERENCE DOCUMENTS | REV | DATE | REVISION DESCRIPTION | DRN | CHK | ENG | APP | EMP |
|---|---|---|---|---|---|---|---|---|---|
| MD-216-49EE-EG-CV-DED-1003 | LAYOUT & DETAILS OF SEWERAGE NETWORK SH-1 & 2 | D01 | 16/02/2023 | AS-BUILT | PML | BJB | APEX | SJM | |
| MD-216-49EE-EG-CV-DED-1002 | LAYOUT & DETAILS OF SEWERAGE NETWORK | C03 | 27/08/2021 | ISSUED FOR CONSTRUCTION AND REVISED AS MARKED △C03 | PML | BJB | APEX | SJM | |
| MD-216-49EE-EG-CV-DED-1001 | LAYOUT & DETAILS OF SEWERAGE NETWORK | C02 | 19/04/2021 | ISSUED FOR CONSTRUCTION AND REVISED AS MARKED △C02 | APEX | BJB | APEX | SJM | |
| MD-216-49EA-EG-CV-GAD-1001 | LAY. & DET. OF STORM WATER DRAINAGE NETWORK | C01 | 06/01/2021 | ISSUED FOR CONSTRUCTION AND REVISED AS MARKED △C01 | APEX | BJB | APEX | SJM | |
| MD-216-1000-EG-GE-PPD-1002 | PROCESS PLANT LAYOUT | A01 | 19/11/2020 | ISSUED FOR REVIEW | APEX | BJB | APEX | SJM | |

| DOCUMENT NUMBER | REFERENCE DOCUMENTS | REV | DATE | REVISION DESCRIPTION | DRN | CHK | ENG | APP | EMP |
|---|---|---|---|---|---|---|---|---|---|

RESERVER FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO

This drawing is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Outotec

The content of these documents are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without prior written permission of Outotec is prohibited.

MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

This document and the information contained herein is Ma'aden proprietary know-how. This document or any part thereof should not be reproduced, stored in a retrieval system or transmitted by any means, electronic or otherwise, or divulged without prior written permission of Ma'aden.

LARSEN & TOUBRO DRAWING NUMBER
019045_045_C_G_24422_002
REVISION 4

OUTOTEC DRAWING NUMBER
900707 __ GAU01 __ AG90__MTL01 __50027
DOCUMENT ID
0U100697790
REVISION 4

| | | |
|---|---|---|
| DRAWN | APEX | 13/11/2020 |
| CHECKED | BJB | 17/11/2020 |
| ENGINEER | APEX | 12/11/2020 |
| APPROVED | SJM | 19/11/2020 |
| EMPLOYER | | |
| SCALE | 1:500 | A0 |

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
PROCESS PLANT SEWERAGE NETWORK
LAYOUT & SCHEDULE FOR OXYGEN PLANT

DRAWING NUMBER
MD - 216 - 49EE - EG - CV - GAD - 1594
REV
D01